Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Project status — the measurement half, built and *validated* (Gate 1)

Kostiantyn Boiar

MSc Dissertation ▪ University of Glasgow

Supervisor: Dr Marwa Mahmoud

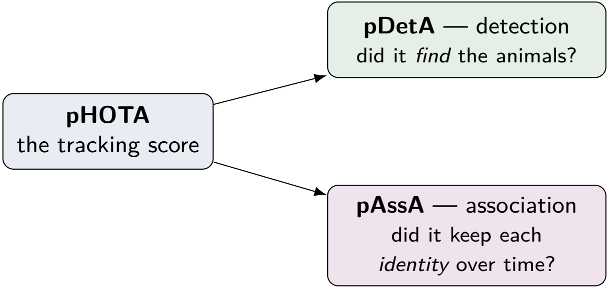
July 15, 2026

The goal, in one sentence

Given a **new animal** in a **new place**, predict how well SAM 3 will track it — *before we run it* — and explain **why**.

- Separately for **finding** the animal (`pDetA`) and **following** it (`pAssA`).
- We are *not* trying to make the tracker score higher — we predict its **reliability**.
- Why it matters: conservation teams point these models at new species/regions constantly, and today **nobody can say in advance** whether to trust them.

What we measure: a promptable tracker, split in two



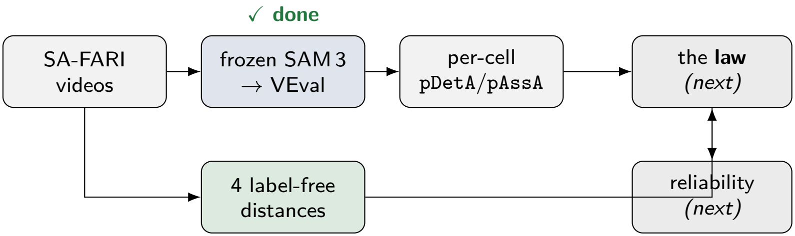
- **SAM 3** is *promptable*: type "`impala`" and it finds and follows every matching animal across a video — with **no training on that species** (zero-shot).
- We analyse **finding** and **following** as two separate numbers — that split is the intellectual core of the project.

The data — SA-FARI

- Thousands of short **camera-trap videos** of wild animals, every frame hand-outlined.
- **99 species** with a full tree-of-life · **741 locations** on 4 continents · a decade of footage (2014–2024).
- Keeps **hard negatives** — a query for an absent animal, where a good model returns nothing (drives the false-positive check).
- Gated outlines from Hugging Face + public frames from Google Cloud, pulled on demand.

species × **place** × **time** — the axes we measure "newness" along.

The plan — and where we are now



- ✓ **Done & validated:** the data pipeline, the four distances, and the frozen-SAM 3 measurement — everything that produces the inputs X and the outputs Y.
- **Next:** fit the small *law* $X \to Y$ and turn it into a reliability estimate (this talk stops before it).

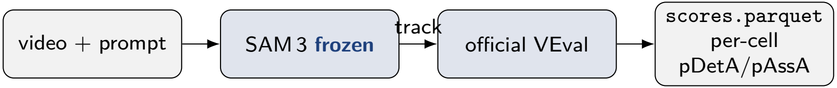
The four “how new is it?” distances (all built, sanity-checked)

- **Taxonomic** — steps up the tree of life to the nearest known species.
- **Visual** — cut the animal out of the frame, “fingerprint” it (DINOv2), compare to known species.
- **Environment** — *erase* the animal, fingerprint the leftover scene, compare to known places; plus a **night/infrared** flag.
- **Temporal** — how many years to the nearest known footage.

The one rule that keeps it honest

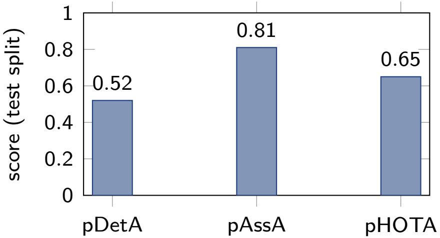
Every distance is computed against the **reference (“seen”) set only** — no information about the target animal can leak in. A test proves a species can never fake “distance 0” to itself.

Measuring SAM 3 — the numbers we predict



- Our harness runs frozen SAM 3 over **every clip** and the **official** scorer turns its outlines into detection/association numbers — one row per **(species, place, year)** cell.
- Built to be **resumable** (a Colab timeout never loses progress); the full test split ran on one A100 in ∼3 hours.

Gate 1 — the measurement is validated



zero-shot SAM 3, full test split

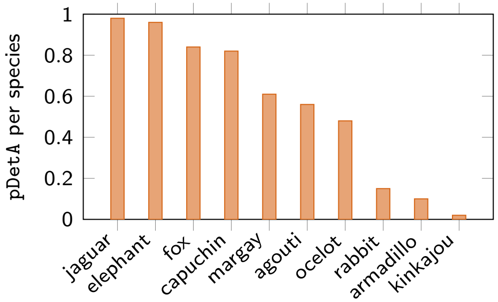
Three independent checks say the numbers are right:

- ✓ **Internally consistent:** pHOTA $\approx \sqrt{\text{pDetA} \cdot \text{pAssA}} = \sqrt{0.52 \cdot 0.81} = 0.65$.
- ✓ **Matches the paper:** our zero-shot 0.65+ the paper's +0.196 fine-tuning gain $\approx$ 0.84 = their *fine-tuned* SAM 3.
- ✓ **Clean on hard negatives:** SAM 3 essentially **never invents** an animal that isn't there.

The whole measurement pipeline reproduces published SAM 3 performance end-to-end.

What the result already shows

- **Finding and following are far apart** (`pDetA` 0.52 vs `pAssA` 0.81): SAM 3 tracks well once it *finds* an animal, but misses about half of them.
- Per species, **finding** spans almost the full range — easy, distinctive animals high; small, cryptic ones low.
- *That spread is the signal* a predictor can learn. If every species scored alike, there would be nothing to predict.



Built to be trusted

- **Config-driven & typed** — an experiment changes a *value*, never the code; a growing automated test suite guards every rule.
- **The leakage firewall** is enforced in code *and* tested — no test-set information can enter a distance, so a "prediction" can never be trivially right.
- **Reproducible on a borrowed GPU** — resumable runs, parallelised downloads, and the *official* evaluator (never re-implemented).

Everything above is committed, tested, and reproduces the published SAM 3 numbers — a solid foundation for the modelling half.

Where we are, and what's next

Done & validated

Data pipeline ✓ four label-free distances ✓ frozen-SAM 3 measurement ✓ **Gate 1 passed** ✓

- **Next:** fit a small **law** — distance in → predicted `pDetA`/`pAssA` out — and check it predicts *held-out* species and places (Gate 2: does out-of-sample signal really exist?).
- Then the deployment tool: a new (species, place) → a prediction + a confidence interval.

Thank you — questions welcome.